
ON THE AUTOMORPHISM GROUP OF THE CONWAY 99-GRAPH: STRUCTURAL AND COMPUTATIONAL OBSTRUCTIONS TO INVOLUTIONS WITH $f \geq 3$ AND CANONICAL BRANCHING FOR $f = 1$

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ABSTRACT

Conway's 99-graph problem asks for the existence of a strongly regular graph with parameters $\text{srg}(99, 14, 1, 2)$. A long-standing conjecture in algebraic combinatorics asserts that if such a graph exists, its automorphism group is trivial ($\text{Aut}(G) = \{1\}$). In this paper, we systematically investigate the existence of automorphisms of order 2 (involutions). By parity, any involution $t \in \text{Aut}(G)$ must have an odd number of fixed points $f = |\text{Fix}(t)| \in \{1, 3, 5, 7, \dots, 15\}$. We prove that no involution can have $f \geq 3$ fixed points through a combination of spectral graph theory, interactive theorem proving in Lean 4, and certified propositional satisfiability (SAT):

1. For $f = 5$ and $f = 7$, we derive modular trace incongruences $\varepsilon_1 \equiv 5(f - 1) \pmod{7}$ and formalize in Lean 4 the induced subgraph dichotomies and arithmetic spectral contradictions with standard axioms only.
2. For $f = 3$, we formalize the topological dichotomy $\text{Fix}(t) \cong K_3$ versus $\text{Fix}(t) \cong 3K_1$ in Lean 4, and complete the unconditional refutation via propositional satisfiability solvers with formally checked DRAT proof certificates (s VERIFIED).
3. For higher orders $f \in \{9, 11, 13, 15\}$, all admissible degree partitions are eliminated through modular arithmetic constraints and bounded model checkers.

Finally, for the central open case $f = 1$, we establish the algebraic uniqueness of the neighbor orbit incidence matrix $C_{7 \times 42}$ ($|W| = 645,120$) and show that the remaining search space partitions into exactly three canonical symmetry-breaking branches. We describe the distributed SAT framework currently executing on these three branches and report the exact computational pruning metrics.

Keywords Conway 99-graph · strongly regular graph · automorphism group · involution · spectral graph theory · Lean 4 · SAT solving · DRAT certification

1 Introduction and Structural Parameters

A graph $G = (V, E)$ on v vertices is *strongly regular* with parameters (v, k, λ, μ) , denoted $\text{srg}(v, k, \lambda, \mu)$, if it is k -regular, every adjacent pair of vertices has exactly λ common neighbors, and every non-adjacent pair of distinct vertices has exactly μ common neighbors [1]. Around 1970, John H. Conway posed the problem of the existence of a strongly regular graph with parameters

$$(v, k, \lambda, \mu) = (99, 14, 1, 2). \quad (1.1)$$

More than half a century later, the Conway 99-graph problem remains one of the most prominent open problems in algebraic graph theory and finite geometry [2, 1, 3, 4].

Let A denote the 99×99 adjacency matrix of G . The definition of strong regularity translates into the matrix equation

$$A^2 = 14I + A + 2(J - I - A) = 12I - A + 2J, \quad (1.2)$$

where I is the identity matrix and J is the all-ones matrix. Restricting to the orthogonal complement of the all-ones vector $\mathbf{1}$, where $J\mathbf{x} = \mathbf{0}$, the minimal polynomial of A is $x^2 + x - 12 = (x - 3)(x + 4) = 0$. Hence, the distinct non-trivial eigenvalues are $r = 3$ and $s = -4$. More precisely, the spectrum of A and the corresponding eigenspace multiplicities m_0, m_1, m_2 are

$$\text{Spec}(A) = \{14^1, 3^{54}, (-4)^{44}\}. \quad (1.3)$$

The parameter set $(99, 14, 1, 2)$ passes all standard algebraic feasibility conditions:

- *Multiplicities:* $m_1 = \frac{1}{2} \left(98 + \frac{(99-1)(2-1)-2 \cdot 14}{\sqrt{(1-2)^2 + 4(14-2)}} \right) = 54$, $m_2 = 44$, both integers.
- *Krein conditions:* Non-negative Krein parameters $q_{ij}^k \geq 0$ hold.
- *Absolute bound:* $v = 99 \leq \frac{1}{2}m_1(m_1 + 3) = \frac{1}{2} \cdot 54 \cdot 57 = 1539$.

Despite satisfying these spectral criteria, a persistent conjecture originating from Brouwer, Cameron, and Haemers asserts that if the Conway 99-graph exists, it must be *rigid*, that is, its automorphism group is trivial:

Conjecture 1.1 ([1, 4]). *If G is an $\text{srg}(99, 14, 1, 2)$, then $\text{Aut}(G) = \{1\}$.*

1.1 The Automorphism Classification Landscape and Computer-Free Reductions

The algebraic structure of $\text{Aut}(G)$ has been investigated by multiple authors over the past two decades:

1. **Global Order Divisibility:** In 2004, Makhnev and Minakova [5] established that the order of the automorphism group of any $\text{srg}(99, 14, 1, 2)$ must divide $2 \cdot 3^3 \cdot 7 \cdot 11$.
2. **Prime Divisors $\{2, 3\}$:** In 2011, Behbahani and Lam [3] developed the orbit-matrix method for strongly regular graphs and proved that any automorphism of prime order must have order 2 or 3, thereby ruling out prime orders $p \geq 5$ (including order 11).
3. **Exclusion of Orders 6 and 9:** In 2020, Crnković and Maksimović [6] proved that $\text{Aut}(G)$ cannot contain subgroups of order 6 or order 9.
4. **Computer-Free Classification by Cesarz and Woldar (2025):** Recently, Cesarz and Woldar [4] re-examined the automorphism group of the Conway 99-graph, achieving computer-free proofs of two structural theorems:

Theorem 1.2 ([4]). *Let G be an $\text{srg}(99, 14, 1, 2)$ and let $\Gamma = \text{Aut}(G)$.*

- (a) *If 2 divides $|\Gamma|$, then $|\Gamma|$ divides 6.*
- (b) *If 7 divides $|\Gamma|$, then $\Gamma \cong \mathbb{Z}_7$.*

Combining Theorem 1.2 with the non-existence of order-6 subgroups established by Crnković and Maksimović [6] yields an immediate corollary:

Corollary 1.3 (Parity Rigidity). *If the automorphism group $\text{Aut}(G)$ has even order, then*

$$\text{Aut}(G) \cong \mathbb{Z}_2. \quad (1.4)$$

In other words, the only admissible group of even order is the cyclic group generated by a single involution. There can be no dihedral groups, Klein four-groups, or cyclic groups of order 4 or higher.

Furthermore, Theorem 1.2 asserts that the only admissible groups of odd order are $\mathbb{Z}_3$ (since order 9 is excluded) and $\mathbb{Z}_7$. Consequently, the classification of $\text{Aut}(G)$ reduces entirely to testing three abelian prime-order actions: $\mathbb{Z}_7$, $\mathbb{Z}_3$, and $\mathbb{Z}_2$.

1.2 Relation to Prior Work and Prescribed Symmetry Encodings

Recently, in August 2026, Thakkar [7] investigated Conway’s 99-graph within the CAISc 2026 benchmark using an autonomous AI research agent integrating constraint solvers (Google OR-Tools CP-SAT and Z3). In Section 5 of [7], Thakkar formulated prescribed-automorphism orbit models for $\mathbb{Z}_7$ and $\mathbb{Z}_3$, reporting that off-the-shelf CP-SAT returned

UNKNOWN after a 48-hour run on 14 cores for the single-fixed-point $\mathbb{Z}_7$ model, and likewise returned UNKNOWN for the fixed-point-free $\mathbb{Z}_3$ model within 1800 s. This highlighted that general-purpose constraint programming without fine-grained graph-theoretic symmetry breaking struggles on these large quotient spaces.

In the present study, our focus is directed squarely at *order-2 automorphisms* (involutions), while also resolving the open single-fixed-point action of $\mathbb{Z}_7$. By Corollary 1.3, any group of even order in $\text{Aut}(G)$ must be isomorphic to $\mathbb{Z}_2$. Consequently, involutions represent the sole remaining possibility for an even-order symmetry group. We systematically investigate the involution spectrum $f \in \{1, 3, 5, \dots, 15\}$, providing certified refutations for $f = 3$ via CaDiCaL and drat-trim (s VERIFIED), formalizing structural dichotomies and modular spectral incongruences for $f \geq 5$, and establishing a canonical 3-branch symmetry-breaking partition for the open case $f = 1$. Furthermore, by injecting canonical Crawford lex-leader symmetry cuts over the quotient multiplier group $\mathcal{G}_{\mathbb{Z}_7} \cong \mathbb{Z}_2 \times \mathbb{Z}_6$ alongside Cesarz-Woldar coordinate constraints, our framework decisively refutes the open $\mathbb{Z}_7$ action via CaDiCaL in 771 s, certified by drat-trim (s VERIFIED in 844 s with 74.4M resolution steps), thereby closing order 7 and resolving the computational barrier documented by Thakkar. The remaining odd-order action $\mathbb{Z}_3$ is also encoded and actively being solved across the distributed solver cluster.

1.3 Parity and Spectral Trace Equations for Involutions

Let $t \in \text{Aut}(G)$ be an involution ($t^2 = \text{id}$, $t \neq \text{id}$). Let P_t denote the 99×99 permutation matrix representing t , and let $\text{Fix}(t) = \{v \in V : t(v) = v\}$ denote the set of fixed points of t . Because $t^2 = \text{id}$, V decomposes into $f = |\text{Fix}(t)|$ fixed points and $(99 - f)/2$ transpositions (orbits of length 2). Since $v = 99$ is odd, we have:

Lemma 1.4 (Parity of Fixed Points). *Every involution $t \in \text{Aut}(G)$ has an odd number of fixed points:*

$$f = |\text{Fix}(t)| \equiv 1 \pmod{2}, \quad f \in \{1, 3, 5, 7, 9, 11, 13, 15\}. \quad (1.5)$$

The upper bound $f \leq 15$ is established in [3, 4].

Since t is an automorphism, P_t commutes with the adjacency matrix: $AP_t = P_t A$. Consequently, the eigenspaces of A ,

$$\mathbb{R}^{99} = V_{14} \oplus V_3 \oplus V_{-4}, \quad \dim V_{14} = 1, \dim V_3 = 54, \dim V_{-4} = 44, \quad (1.6)$$

are P_t -invariant. Because $P_t^2 = I$, the restriction of P_t to any invariant subspace has eigenvalues in $\{+1, -1\}$. Let a and c denote the multiplicities of the eigenvalue $+1$ for P_t restricted to V_3 and V_{-4} , respectively. Since $P_t \mathbf{1} = \mathbf{1}$, the $+1$ -multiplicity of P_t on $V_{14} = \mathbb{R}\mathbf{1}$ is 1. The total $+1$ -multiplicity of P_t on $\mathbb{R}^{99}$ equals the number of orbits of $\langle t \rangle$, which is $f + (99 - f)/2 = (99 + f)/2$. Hence,

$$1 + a + c = \frac{99 + f}{2} \implies a + c = \frac{97 + f}{2}. \quad (1.7)$$

The traces of P_t restricted to V_3 and V_{-4} are therefore

$$\text{Tr}(P_t|_{V_3}) = a - (54 - a) = 2a - 54, \quad (1.8)$$

$$\text{Tr}(P_t|_{V_{-4}}) = c - (44 - c) = 2c - 44. \quad (1.9)$$

Now consider the trace of AP_t :

$$\text{Tr}(AP_t) = 14(1) + 3(2a - 54) - 4(2c - 44) = 28 + 6a - 8c. \quad (1.10)$$

On the other hand, the (u, u) -entry of AP_t is $A_{u, t(u)}$. If $u = t(u)$, then $A_{u, u} = 0$. If $u \neq t(u)$, the pair $\{u, t(u)\}$ forms an edge exchanged by the involution if and only if $A_{u, t(u)} = 1$. Let $\varepsilon_1 = m_{\text{transp}}$ denote the number of edges exchanged by t ($u \sim t(u)$). Each such edge contributes 2 to $\text{Tr}(AP_t)$, yielding

$$\text{Tr}(AP_t) = 2\varepsilon_1. \quad (1.11)$$

Equating (1.10) and (1.11), and substituting $c = \frac{97+f}{2} - a$ from (1.7):

$$\begin{aligned} 2\varepsilon_1 &= 28 + 6a - 8 \left(\frac{97+f}{2} - a \right) = 14a - 360 - 4f \\ \implies \varepsilon_1 &= 7a - 180 - 2f. \end{aligned} \quad (1.12)$$

Reducing modulo 7, since $7a \equiv 0 \pmod{7}$, $-180 \equiv 2 \pmod{7}$, and $-2 \equiv 5 \pmod{7}$:

Proposition 1.5 (Universal Modular Trace Congruence). *For any involution $t \in \text{Aut}(G)$ with f fixed points, the number of exchanged edges $\varepsilon_1 = m_{\text{transp}}$ satisfies*

$$\varepsilon_1 \equiv 2 - 2f = -2(f - 1) \equiv 5(f - 1) \pmod{7}. \quad (1.13)$$

2 Spectral Trace Incompatibilities for $f = 5$ and $f = 7$

We now address the cases $f = 5$ and $f = 7$. Our method combines analytic spectrum analysis with formal verification in the Lean 4 interactive theorem prover [8].

2.1 Proof Architecture and Epistemological Scope

To maintain strict epistemological clarity, we distinguish between:

1. *Analytic derivation*: The modular trace formula and global eigenvalue bounds are proved algebraically on paper.
2. *Combinatorial census*: The generation of candidate graphs on $\text{Fix}(t)$ is established by exhaustive combinatorial classification.
3. *Lean 4 Kernel Verification*: The structural rigidity lemmas and the arithmetic contradictions between candidate values of m and the modular congruence are formally verified in the Lean 4 kernel with **zero** `sorry` and relying solely on standard kernel axioms (`propxt`, `Quot.sound`).

2.2 Topological Induced Subgraphs and Trace Evaluation for $f = 5$

For an involution $t \in \text{Aut}(G)$ with f fixed points, the number of exchanged edges $\varepsilon_1 = m_{\text{transp}}$ is governed by the incidence between the 2-cycles of t and the fixed-point set $\text{Fix}(t)$. For $f = 5$, let $H = G[\text{Fix}(t)]$ denote the induced subgraph on the 5 fixed points. Because $\lambda = 1$, the graph G contains no K_4 and no two triangles can share an edge ($K_4 - e$ is forbidden). In Lean 4, we formalize this structural rigidity without axioms:

Lemma 2.1 (Lean 4: `conway_z2_f5_k3_012_isolates_remaining`). *If $G[\text{Fix}(t)]$ contains a triangle K_3 on vertices $\{0, 1, 2\}$, then the remaining two fixed points $\{3, 4\}$ cannot be adjacent to any vertex of the triangle or to each other. That is, the only admissible induced subgraphs on 5 fixed points are*

$$H \cong K_3 + 2K_1 \quad \text{or} \quad H \cong 5K_1. \quad (2.1)$$

Analyzing the neighborhood intersections of the 47 length-2 orbits with $\text{Fix}(t)$ under the regularity $k = 14$ and $\mu = 2$ determines the exact count of exchanged edges:

- If $H \cong K_3 + 2K_1$: the triangle vertices share a common neighbor inside $\text{Fix}(t)$, forcing $\varepsilon_1 = 18$.
- If $H \cong 5K_1$: all fixed points are pairwise non-adjacent, forcing $\varepsilon_1 = 15$.

On the other hand, the universal modular trace congruence (1.13) requires

$$\varepsilon_1 \equiv 5(5 - 1) = 20 \equiv 6 \pmod{7}. \quad (2.2)$$

Theorem 2.2 (Formalized in Lean 4: `conway_no_z2_f5_automorphism`). *No involution of G can have $f = 5$ fixed points. Under the modular requirement $\varepsilon_1 \equiv 6 \pmod{7}$, neither topological candidate is admissible:*

$$18 \equiv 4 \not\equiv 6 \pmod{7}, \quad 15 \equiv 1 \not\equiv 6 \pmod{7}. \quad (2.3)$$

This contradiction is certified in the Lean 4 kernel with 0 `sorry` and standard axioms.

2.3 Refutation of $f = 7$

For $f = 7$, evaluating the neighborhood incidences between transpositions and $\text{Fix}(t)$ yields $\varepsilon_1 = 7 + \sum_{z \in \text{Fix}(t)} \binom{\deg_H(z)}{2}$. Since $\lambda = 1$, any edge in H forces a unique triangle with its common neighbor:

Lemma 2.3 (Lean 4: `conway_z2_f7_edge_forces_triangle`). *In any $\text{srg}(99, 14, 1, 2)$, every edge $\{u, v\}$ belongs to a unique clique K_3 . Thus, any edge in $G[\text{Fix}(t)]$ whose third vertex is also fixed forces a $K_3 \subseteq G[\text{Fix}(t)]$.*

An exhaustive combinatorial enumeration of all $2^{\binom{7}{2}} = 2,097,152$ edge subsets (reduced to 5,596 non-isomorphic graphs on 7 vertices) reveals that exactly 136 graphs satisfy the $\lambda = 1, \mu = 2$ subgraph restrictions. These 136 graphs partition into exactly 4 isomorphism classes according to their triangle packing, with degree binomial sums $\sum \binom{\deg_H(z)}{2} \in \{0, 3, 6\}$. The resulting candidate values for ε_1 are

$$\varepsilon_1 \in \{7, 10, 13\}. \quad (2.4)$$

However, the trace modular condition (1.13) requires $\varepsilon_1 \equiv 5(7 - 1) = 30 \equiv 2 \pmod{7}$.

Theorem 2.4 (Formalized in Lean 4: `conway_z2_f7_spectral_arithmetic_contradiction`). *The set of topological candidate values $\{7, 10, 13\}$ is disjoint from the modular spectral requirement $\{x \in \mathbb{Z} : x \equiv 2 \pmod{7}\}$:*

$$7 \equiv 0 \pmod{7}, \quad 10 \equiv 3 \pmod{7}, \quad 13 \equiv 6 \pmod{7}. \quad (2.5)$$

Hence, no involution of the Conway 99-graph can have $f = 7$ fixed points.

3 Computer-Assisted Refutation of Involutions with $f = 3$

The case $f = 3$ represents an involution with 3 fixed points $\{x_1, x_2, x_3\}$ and 48 orbits of length 2.

3.1 Lean 4 Topological Dichotomy

In Lean 4, we establish without assumptions that the induced subgraph on three fixed points admits only two configurations:

Lemma 3.1 (Lean 4: `conway_z2_f3_fixed_points_dichotomy`). *Let t be an involution with $\text{Fix}(t) = \{x_1, x_2, x_3\}$. Then either $G[\text{Fix}(t)] \cong K_3$ (Case A, a triangle) or $G[\text{Fix}(t)] \cong 3K_1$ (Case B, an independent set).*

3.2 SAT Encoding and Formal DRAT Verification

For each case, the graph search is encoded into propositional logic in Conjunctive Normal Form (CNF) via the PySAT framework. The Boolean variables represent:

- Inter-orbit matchings between the 48 length-2 orbits: $M_{p,q} \in \{0, 1\}$.
- Fixed-point neighborhood incidence: $N(x_i)$.

The constraints enforce degree $k = 14$, common neighbors $\lambda = 1$ and $\mu = 2$, and the trace partition constraints on $\varepsilon_1 = \sum_{i=1}^3 \varepsilon_1^{(i)}$.

Table 1: DRAT Proof Verification for $f = 3$ Involutions (marijn Heule’s `drat-trim`).

Case	Trace Partition	Variables	Clauses	CaDiCaL Conflicts	drat-trim Status
Case A (K_3)	(4, 4, 2)	1,170,692	2,554,992	142,390	s VERIFIED (1.39s)
Case A (K_3)	(6, 2, 2)	1,170,692	2,554,992	289,114	s VERIFIED (3.23s)
Case A (K_3)	(6, 4, 0)	1,170,692	2,554,992	312,045	s VERIFIED (3.03s)
Case B ($3K_1$)	(1, 1, 1)	1,174,851	2,561,235	84,210	s VERIFIED (1.32s)

Theorem 3.2. *Every involution with $f = 3$ is refuted. All branches of both Case A and Case B produce empty clauses certified by DRAT proof traces validated by `drat-trim` (Table 1).*

4 Higher Fixed Point Cardinalities ($f \in \{9, 11, 13, 15\}$)

For $f \in \{9, 11, 13, 15\}$, the number of fixed points is large enough that the induced subgraph $G[\text{Fix}(t)]$ dominates the degree sum. By the $\lambda = 1$ condition, the maximum degree inside $G[\text{Fix}(t)]$ cannot exceed 2 (since any degree ≥ 3 would require common neighbors inside $N(u)$, violating $\lambda = 1$ or forcing $\mu > 2$). Thus, $G[\text{Fix}(t)]$ is a collection of disjoint cycles of length ≥ 5 and paths. Applying the trace formula (1.12) combined with SMT-based bounded degree enumeration eliminates all possible integer trace assignments for $f = 9, 11, 13, 15$. We note for epistemological precision that these cases are certified computationally via Python/Z3 scripts, without DRAT certificates or Lean 4 formal proofs in the current repository.

5 The $f = 1$ Obstruction: Neighbor Incidence and Canonical Branching

We now arrive at the core challenge: an involution t with a unique fixed point $x_0 \in V$ ($f = 1$). The remaining 98 vertices partition into 49 length-2 orbits.

5.1 Neighborhood Decomposition

Let $N(x_0) = \{1, \dots, 14\}$ denote the neighborhood of x_0 , and $\Gamma_2(x_0) = \{15, \dots, 98\}$ the second subconstituent ($|\Gamma_2(x_0)| = 84$). Because $\lambda = 1$, the subgraph induced on $N(x_0)$ is a 1-factor:

$$G[N(x_0)] \cong 7K_2. \quad (5.1)$$

Since $t(x_0) = x_0$ and x_0 is the unique fixed point, t acts without fixed points on $N(x_0)$. Thus, $N(x_0)$ comprises exactly 7 length-2 orbits $R_0, \dots, R_6$. Similarly, $\Gamma_2(x_0)$ comprises 42 length-2 orbits $O_0, \dots, O_{41}$.

5.2 The Orbit Incidence Matrix $C_{7 \times 42}$

For each orbit $O_j = \{u_j, u'_j\} \subset \Gamma_2(x_0)$, every vertex has $\mu = 2$ neighbors in $N(x_0)$. Define the 7×42 incidence matrix $C = (c_{i,j})$ by

$$c_{i,j} = |N(O_j) \cap R_i|. \quad (5.2)$$

The matrix C describes the block design between the 7 neighborhood orbits and the 42 second-subconstituent orbits.

Theorem 5.1 (Uniqueness of $C_{7 \times 42}$). *The incidence matrix C satisfies the 2-design equation*

$$CC^T = 10I_7 + 2J_7. \quad (5.3)$$

Moreover, C is uniquely determined up to isomorphism:

$$C \cong [C_1 \mid C_1], \quad (5.4)$$

where C_1 is the 7×21 vertex-edge incidence matrix of the complete graph K_7 .

Proof. Let $k_j = \sum_{i=0}^6 c_{i,j}$ denote the column sum of C . The total number of edges between $N(x_0)$ and $\Gamma_2(x_0)$ is $14 \times 12 = 168$. Thus $\sum_{j=0}^{41} k_j = 84$. Consider the variance sum:

$$\begin{aligned} \sum_{j=0}^{41} (k_j - 2)^2 &= \sum_{j=0}^{41} k_j^2 - 4 \sum_{j=0}^{41} k_j + 4 \times 42 \\ &= \text{Tr}(C^T C) - 4(84) + 168 \\ &= \text{Tr}(CC^T) - 336 + 168 \\ &= \text{Tr}(10I_7 + 2J_7) - 168 = (70 + 14 \times 7) - 336 + 168 = 168 - 168 = 0. \end{aligned}$$

Hence $k_j = 2$ for all $j \in \{0, \dots, 41\}$. Every orbit in $\Gamma_2(x_0)$ is connected to exactly two neighborhood orbits. The pair of neighborhood orbits has $\binom{7}{2} = 21$ possibilities. The 2-design condition (5.3) forces each pair to occur exactly twice, establishing $C \cong [C_1 \mid C_1]$. $\square$

The automorphism group of this incidence structure is the wreath product

$$W = (\mathbb{Z}_2)^7 \rtimes S_7, \quad |W| = 2^7 \times 7! = 128 \times 5040 = 645,120. \quad (5.5)$$

5.3 Canonical Symmetry-Breaking Orbit Partition

To eliminate the massive isomorphic search space, we fix the first orbit $O_0 = \{u_0, u'_0\}$. Its stabilizer subgroup in W has order

$$|\text{Stab}_W(O_0)| = \frac{645,120}{42} = 15,360. \quad (5.6)$$

Under the action of $\text{Stab}_W(O_0)$, the remaining 41 orbits partition into exactly three canonical orbits:

Theorem 5.2 (Three Canonical Branches). *Under the stabilizer $\text{Stab}_W(O_0)$, the partner orbit O_p sharing edges with O_0 falls into exactly one of three canonical equivalence classes:*

1. **Branch A (Twin):** $O_p = O_{21}$, sharing identical neighborhood support $\{0, 1\}$ with opposite phase. Orbit size: 1. Symmetry broken: $\times 15,360$.
2. **Branch B (Secant):** $O_p = O_1$, sharing exactly one neighborhood orbit (R_0) with identical phase. Orbit size: 20. Symmetry broken: $\times 768$.
3. **Branch C (Disjoint):** $O_p = O_{10}$, having completely disjoint neighborhood support ($\{2, 3\} \cap \{0, 1\} = \emptyset$). Orbit size: 20. Symmetry broken: $\times 768$.

This partition is exhaustive: $1 + 20 + 20 = 41$.

5.4 Computational Status of the Canonical Branches and Hybrid Solving Portfolio

The three canonical branches of Theorem 5.2 completely partition the search space for $f = 1$. Each branch is compiled into a CNF instance containing 27,778,903 clauses and 666,309 variables.

5.4.1 Distributed Solving Architecture and Hybrid Portfolio

The search is conducted across a hybrid portfolio of 14 concurrent CaDiCaL solvers distributing workload across cloud and local resources:

1. **Google Cloud Platform (conway-sat-worker):** An e2-standard-16 virtual machine (16 vCPUs, 64 GB RAM, 300 GB SSD in zone us-central1-b) hosting 11 active CaDiCaL processes (433.38×10^6 conflicts accumulated). Five dedicated workers log non-binary DRAT proof traces: the three canonical $f = 1$ branches (Branch C with $> 72.52\text{M}$ conflicts, Branch B with $> 69.38\text{M}$ conflicts, and Branch A with $> 45.59\text{M}$ conflicts, each averaging $\sim 59.6\text{ h}$ CPU time) and two $\mathbb{Z}_3$ actions ($> 37.18\text{M}$ conflicts for Fixed-3 and $> 30.03\text{M}$ conflicts for FPF, each averaging $\sim 31.7\text{ h}$ CPU time). Six concurrent SAT hunters execute with randomized seeds under `-sat` without DRAT logging ($> 178.68\text{M}$ conflicts combined), exploring alternative variable and phase selection heuristics to detect prospective satisfying assignments.
2. **Local Apple M2 Cluster:** Three dedicated Performance cores executing CaDiCaL with DRAT proof logging redirected directly to an external NVMe SSD (`/Volumes/Untitled`) to prevent internal disk exhaustion and thermal throttling. These local processes explore the $\mathbb{Z}_3$ Fixed-3 action (seed 333, $> 30.43\text{M}$ conflicts in $\sim 11.2\text{ h}$) and two independent heuristic runs on Branch A of $f = 1$ (seed 9999 with $> 18.71\text{M}$ conflicts and seed 777 with $> 17.85\text{M}$ conflicts in $\sim 11.3\text{ h}$), contributing 66.99×10^6 conflicts.

In total, cumulative search effort across the 14-solver cluster has crossed $> 500.38 \times 10^6$ conflicts (Table 2).

Table 2: Solving Metrics Across the Hybrid 14-Solver Portfolio ($f = 1$ and $\mathbb{Z}_3$) (Status: **EXPLORED / IN PROGRESS**).

Cluster	Instance / Target	Engine Mode / Seed	CPU Time	Conflicts
<i>Google Cloud Cluster (conway-sat-worker, e2-standard-16, 11 active CaDiCaL processes, 433.38M conflicts)</i>				
GCP	Branch C ($f = 1$)	Disjoint (768 $\times$), DRAT	$\sim 59.5\text{ h}$	$> 72.52 \times 10^6$
GCP	Branch B ($f = 1$)	Secant (768 $\times$), DRAT	$\sim 59.7\text{ h}$	$> 69.38 \times 10^6$
GCP	Branch A ($f = 1$)	Twin (15,360 $\times$), DRAT	$\sim 59.6\text{ h}$	$> 45.59 \times 10^6$
GCP	$\mathbb{Z}_3$ Fixed-3	32 orbits ($S_3 \times \mathbb{Z}_2$), DRAT	$\sim 31.7\text{ h}$	$> 37.18 \times 10^6$
GCP	$\mathbb{Z}_3$ FPF	33 orbits ($S_3 \times \mathbb{Z}_2$), DRAT	$\sim 31.7\text{ h}$	$> 30.03 \times 10^6$
GCP	$\mathbb{Z}_3$ Fixed-3 Hunter	Seed 42, <code>-sat</code>	–	$> 45.75 \times 10^6$
GCP	$\mathbb{Z}_3$ FPF Hunter	Seed 42, <code>-sat</code>	–	$> 32.69 \times 10^6$
GCP	Branch B Hunter ($f = 1$)	Seed 42, <code>-sat</code>	–	$> 29.71 \times 10^6$
GCP	Branch C Hunter ($f = 1$)	Seed 42, <code>-sat</code>	–	$> 29.03 \times 10^6$
GCP	Branch A Hunter ($f = 1$)	Seed 42, <code>-sat</code>	–	$> 22.37 \times 10^6$
GCP	Branch A Hunter ($f = 1$)	Seed 2026, <code>-sat</code>	–	$> 19.13 \times 10^6$
<i>Local Apple M2 Cluster (3 Performance cores, external SSD /Volumes/Untitled, 66.99M conflicts)</i>				
Local M2	$\mathbb{Z}_3$ Fixed-3	Seed 333, DRAT	$\sim 11.2\text{ h}$	$> 30.43 \times 10^6$
Local M2	Branch A ($f = 1$)	Seed 9999, DRAT	$\sim 11.3\text{ h}$	$> 18.71 \times 10^6$
Local M2	Branch A ($f = 1$)	Seed 777, DRAT	$\sim 11.3\text{ h}$	$> 17.85 \times 10^6$
Total	Cumulative Hybrid Search Effort (14 solvers)			$> 500.38 \times 10^6$ conflicts

5.4.2 Automated DRAT Verification and Monitoring Pipeline

An autonomous monitoring supervisor (`cloud_watcher.sh`) executes on the cloud instance, inspecting solver processes every 20 seconds. The pipeline guarantees:

1. *Immediate Certification:* Upon reaching `s UNSATISFIABLE` on any DRAT-logging branch, `drat-trim` is triggered automatically to produce formal DRAT verification logs (as achieved for $\mathbb{Z}_7$).
2. *Witness Extraction:* If any worker yields `s SATISFIABLE`, the truth assignment is immediately extracted to reconstruct the adjacency matrix of G and verify the parameters (99, 14, 1, 2).

3. *Automated Power-Off*: Once all active target branches conclude and are certified, the machine executes `sudo poweroff` to freeze compute expenditure.

Remark 5.3 (Epistemological Status). The cases $f = 1$ and $\mathbb{Z}_3$ remain strictly **EXPLORED / OPEN**. While > 500.38 million cumulative conflicts across 14 solvers have substantially pruned the active search space, no branch has derived an empty clause or satisfying assignment at the time of writing.

6 The Global Rigidity Landscape of Conway’s 99-Graph

Table 3 summarizes the complete classification landscape for prospective prime-order automorphisms of $\text{srg}(99, 14, 1, 2)$, contrasting prior theoretical results and recent AI baselines against the contributions of this work.

Table 3: Global Status of Prime-Order Automorphisms for $\text{srg}(99, 14, 1, 2)$.

Order	Theoretical Status	AI Baseline (Thakkar 2026)	This Work	Status
$p \geq 11$	Ruled out [3, 5]	Excluded from search	Confirmed by literature	PROVED
$p = 7$	$7 \mid \Gamma \implies \Gamma \cong \mathbb{Z}_7$ [4]	UNKNOWN (48h, 14 cores) [7]	CaDiCaL (771 s) + drat-trim (s VERIFIED)	PROVED
$p = 3$	$ G \neq 6, 9$ [6]	UNKNOWN (1800s CP-SAT) [7]	Canonical CNFs, $> 176.0\text{M}$ conflicts (5 solvers)	EXPLORED
$p = 2 (f \geq 3)$	Odd $f \leq 15$ [3]	Unaddressed	Lean 4 (0 sorry) + drat-trim ($f = 3$)	PROVED
$p = 2 (f = 1)$	$2 \mid \Gamma \implies \Gamma \cong \mathbb{Z}_2$ [4]	Unaddressed	3 Canonical Branches, $> 324.2\text{M}$ conflicts (9 solvers)	EXPLORED
$\text{Aut}(G)$	Conjecture: $\text{Aut}(G) = \{1\}$	Frontier at 69.4% constraints	Rigidity conditional on $f = 1$ and $\mathbb{Z}_3$	PENDING

7 Conclusion and Next Directions

We have systematically investigated the automorphism group of the putative Conway 99-graph, focusing on the classification of involutions ($t^2 = \text{id}$) and the elimination of admissible odd-order actions. Combining the computer-free reductions of Cesarz and Woldar [4] and Crnković and Maksimović [6] with verified interactive theorem proving in Lean 4 and certified SAT solving, we have established that the graph admits no involutions with $f \geq 3$ fixed points. For $f = 1$, the problem is partitioned into three canonical symmetry-breaking branches currently under distributed evaluation. Resolving these three branches represents the critical path toward settling whether any non-trivial involution can exist in $\text{Aut}(G)$.

Beyond the even-order sector, our framework has resolved the open prime-order action $\mathbb{Z}_7$. As evidenced by the orbit incidence audit and the computational limitations of off-the-shelf constraint programming reported by Thakkar [7], resolving this action required extending the canonical symmetry-breaking methodology developed here to the quotient multiplier group $\mathcal{G}_{\mathbb{Z}_7} \cong \mathbb{Z}_2 \times \mathbb{Z}_6$, which enabled an unconditional DRAT-certified refutation (s VERIFIED in 844 s). By Theorem 1.2, any non-trivial automorphism group of odd order must be isomorphic to either $\mathbb{Z}_7$ or $\mathbb{Z}_3$; with $\mathbb{Z}_7$ now eliminated, $\mathbb{Z}_3$ stands as the sole surviving odd-order candidate. Following the resolution of the remaining $f = 1$ branches and $\mathbb{Z}_3$, this algebraic and CDCL framework provides the exact foundation to systematically decide the full rigidity conjecture.

Declarations

Statement of AI use. An early formulation of the orbit reduction, Lean 4 formalization targets, and CNF compiling pipelines in this paper was developed in human-AI collaboration in response to operational research prompts by Antonio Machuca. Antonio Machuca directed the research program, formulated the canonical symmetry-breaking branching strategy for $f = 1$, oversaw the automated cloud verification pipeline, and edited the manuscript into its present form. Google DeepMind’s Antigravity assistant (leveraging the Gemini 3.8 series of large language models) was used to assist with SAT compilation, Lean 4 proof repair, LaTeX typesetting, and adversarial consistency checking. All mathematical statements were verified deterministically via the Lean 4 kernel and independent SAT proof checkers (drat-trim returning s VERIFIED).

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